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Lab Exercises:

Complete the following exercises in order. For each exercise, show your code, the output, and a brief explanation of what the result means.

Part 1 Exercises

Exercise 1 — Change the Goal City

Run the program again.

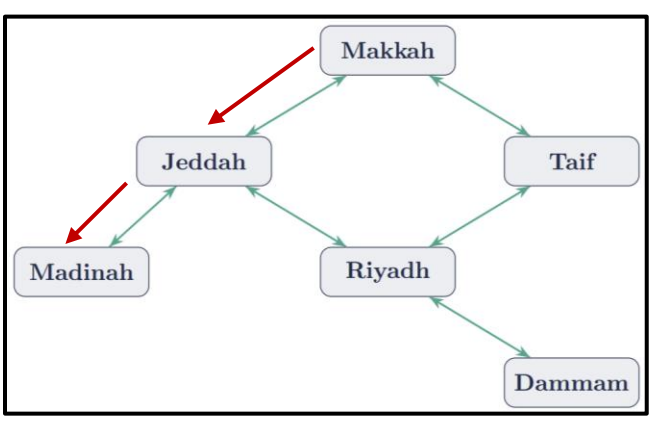
```
🔍 Searching from 'Makkah' to 'Madinah' using BFS...

  Step 1: Visiting 'Makkah'
  Step 2: Visiting 'Jeddah'
  Step 3: Visiting 'Taif'
  Step 4: Visiting 'Riyadh'
  Step 5: Visiting 'Madinah'

✅ Goal reached! 'Madinah' found.

📍 Path found (2 steps):
Makkah → Jeddah → Madinah
```

Answer these questions in your lab report:

a. How many steps did BFS take to reach Madinah? **5 Steps**

b. What is the path it found? **Makkah → Jeddah → Madinah**

c. Is this the shortest path? Verify by looking at the city map diagram.
Yes, it's



Exercise 2 – Add a New City

Add a new city Abha to the map, connected to Taif:

a. In city map, add: "Abha": ["Taif"]

b. Also update Taif's list to include Abha.

c. Set goal city = "Abha" and run.

```
🔍 Searching from 'Makkah' to 'Abha' using BFS...

  Step 1: Visiting 'Makkah'
  Step 2: Visiting 'Jeddah'
  Step 3: Visiting 'Taif'
  Step 4: Visiting 'Riyadh'
  Step 5: Visiting 'Madinah'
  Step 6: Visiting 'Abha'

✅ Goal reached! 'Abha' found.

📍 Path found (2 steps):
Makkah → Taif → Abha
```

d. Write the path BFS found from Makkah to Abha. **Makkah → Taif → Abha**

Part 2 Exercises

Exercise 3 – Add a New Clue

Add the following new clue to the KB (after the existing kb.tell lines):

```
nora_key = Symbol ("NoraHasKey")
kb.tell ( nora_key ) # Nora has a key (new evidence )
```

Also add "NoraHasKey" to all symbols.

Then add a new query to find out if Nora is guilty:

```
nora_guilty = Symbol ("NoraIsGuilty")
answer3 = check_all ( kb, nora_guilty, all_symbols )
print ( f"Is Nora guilty ? { answer3 }")
```

Run the program and answer:

```
====================================================
  CS3081 Lab 2 – Knowledge Base Detective
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🔍 Query:  Is Lina guilty?
✅ YES  –  The KB ENTAILS that Lina is guilty.

🔍 Query:  Does Sara have a key?
❌ NO   –  The KB does NOT entail Sara has a key.

Is Nora guilty ? False
```

a. Does the KB entail Nora is guilty? Why or why not?

No, KB doesn't entail Nora is guilty, because the rule says (If Lina has a key → Lina is guilty)

b. What extra clue would you need to add to prove Nora is guilty? Write the kb.tell line.

Kb.tell(implication(nora_key, nora_guilty))

Exercise 4 – Remove a Clue

Try commenting out (adding # at the start of) Clue 3 (kb.tell(lina seen)):

kb.tell (lina_seen) <-- commented out

Run the program again and answer:

```
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  CS3081 Lab 2 – Knowledge Base Detective
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🔍 Query:  Is Lina guilty?
❌ NO   –  The KB does NOT entail that Lina is guilty.

🔍 Query:  Does Sara have a key?
❌ NO   –  The KB does NOT entail Sara has a key.

Is Nora guilty ? False
```

a. Is Lina still guilty according to the KB? What changed?

No, it's not guilty.

b. Explain in your own words why removing that one clue changed the result.

Because the rule was (If Lina was seen near the room → Lina has a key) and (If Lina has a key → Lina is guilty). But now we commented kb.tell (lina_seen), so the rule is broken.

Discussion Questions

Answer these questions in your lab report. Full sentences, please

Search Questions

1. In Part 1, BFS visited Madinah even though it is not on the path to Dammam. Why does BFS visit "wrong" cities, and is this a problem?

No, it's not a problem. BFS is using queue (FIFO) which explore level by level to find the shortest path

2. Compare BFS to DFS (from Lab 1). For finding the shortest path, which is better and why?

BFS is using queue (FIFO) and it explore level by level which guarantee to find the shortest bath.

While DFS it using stack (LIFO) and explores the deepest node, and it doesn't guarantee to find the shortest path.

3. Imagine you added road distances (e.g., Jeddah to Riyadh = 900 km). BFS does not consider distances – it just counts hops. What kind of search algorithm could use those distances to find a cheaper route?

I will use A* because it considers both the path cost and a heuristic

Knowledge Questions

4. In your own words, what is the difference between implication (P → Q) and entailment (KB |= α)?

implication (P → Q): It's a rule that states if P is true, then Q must also be true. It is false only when P is true and Q is false.

(KB |= α): It will check the all model and return True only if the KB was True and the α is also True. Then we can say that KB entail α.

5. In Part 2, model checking tried all $2^4 = 16$ worlds. If we had 10 symbols instead of 4, how many worlds would need to be checked? Is this practical for very large problems?

The rule is 2^n, n = 10 so $2^{10} = 1024$ worlds

No, it's not practical